COMPONENT SELECTION & DATASHEET ENGINEERING ANALYSIS

Technical Justification & Parameter Verification of Active Power ICs

1. TPS2121RUXR - Seamless Dual Power Multiplexer

Function: Automatic selection between External Vehicle 5V (Priority 1) and USB-C 5V (Priority 2).

- Input Range: 2.7V to 22V (Absolute max 24V). Ideal for 5V automotive environments.

- Switchover Time: 5 microseconds fast switchover ensures zero interruption on VBUS_MUX.

- Reverse Current Blocking: Dual back-to-back N-FET architecture prevents back-feeding from USB-C into the vehicle DC-DC converter or vice-versa.

- Threshold Configuration: External resistor divider R1/R3 sets OV1 cutoff to 5.80V; R4/R6 sets OV2 cutoff to 5.80V. PR1 divider selects Priority 1 when Vehicle 5V > 4.40V.

2. BQ25895RTWR - 1S Li-ion Fast Charger & NVDC Power Path

Function: Single-cell battery management, system power path regulation, and JEITA temperature supervision.

- NVDC Architecture: System rail (SYS) is regulated independently from battery voltage. When battery is depleted or absent, SYS remains stable at 3.65V from external 5V input.

- Fast Charge Capability: Up to 5.0A charge current with 93% buck switching efficiency.

- Autonomous Operation: Hardwired ILIM resistor R8 (158 Ohm) sets autonomous 3.25A input current limit even before ESP32 firmware initializes I2C registers.

- JEITA Compliance: Thermistor divider R9 (5.23k) / R10 (30.1k) dynamically suspends charging outside safe battery temperatures (0°C to 45°C/60°C).

3. TPS61088RHLR - 10A Synchronous Main Boost Converter

Function: Stepping up battery/NVDC rail (3.0V to 4.2V) to regulated 5V_SYS (3.5A continuous / 5A peak).

- Integrated Low-RDSon FETs: 13 mOhm low-side switch and 21 mOhm high-side synchronous rectifier.

- Peak Current Limit: 10A cycle-by-cycle inductor current limit set by R21 (12k).

- Low Noise PWM: MODE pin tied to GND enforces forced PWM switching at 600kHz, eliminating audio-band subharmonic noise and providing tight transient regulation.

4. TPS62162DSGR - 3.3V Step-Down Converter

Function: Generating ultra-stable 3V3_SYS peripheral rail from 5V_SYS.

- DCS-Control Topology: Combines hysteretic and voltage-mode control for immediate transient response.

- Fixed Output: Factory-trimmed 3.30V internal divider eliminates external resistor tolerances and drift.

- High Frequency: 2.25MHz switching permits compact 2.2uH SMD inductor and low ripple (< 10mV).

5. TPS62130RGTR & AO3401A - Dual Cellular Supply Options

Function: Providing verified dual hardware paths for physical A7682S breakout.

- Option A (AO3401A): SOT-23 P-MOSFET switch passes 5V_SYS directly to SIM_PWR_A with < 0.09V drop.

- Option B (TPS62130): 3.0A synchronous buck converter steps down 5V_SYS to 3.80V (SIM_PWR_B).

- Jumper JP1: Allows instant, irreversible hardware selection after physical bench voltage verification.

6. Battery Protection: DW01A-G + Dual FS8205A

Function: Dedicated secondary hardware safety barrier against overcharge, overdischarge, and short-circuits.

- Dual Paralleled TSSOP-8: Paralleling two FS8205A ICs doubles continuous current rating to 6.0A and halves on-resistance to 12.5 mOhm, eliminating thermal hotspots under 5A peak boost draw.